

# Simulation-Based Safety Assessment of Building Evacuation Using Floor-Plan-Aware Crowd Modeling

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**Abstract**—This paper presents PeopleFlow, a floor-plan-aware evacuation simulation framework designed to support simulation-based safety assessment of buildings under emergency conditions. The system converts architectural floor plans into simulation-ready environments, models pedestrian movement and route-selection behavior, and produces interpretable safety indicators including total evacuation time, congestion hotspots, exit utilization, and hazard-exposure-aware outcomes. Rather than treating evacuation analysis as a single deterministic estimate, the workflow evaluates safety through structured comparative scenarios including normal evacuation, blocked-exit conditions, increased occupancy, and guided routing strategies. The platform integrates reproducible experiment execution, replayable simulation sessions, and artifact-backed reporting to support traceable engineering analysis. We demonstrate the approach on multiple building-layout case studies and show how scenario comparisons reveal unsafe bottlenecks, overloaded exits, and policy-dependent differences in evacuation performance. We further describe a calibration and validation workflow based on empirical evacuation targets and density-speed consistency checks to strengthen interpretability of the simulation results. The study positions simulation not as a guarantee of safety, but as a structured decision-support tool for identifying safer configurations, prioritizing interventions, and communicating evacuation risk using measurable evidence.

**Index Terms**—Evacuation simulation, crowd modeling, building safety, floor plan analysis, emergency planning, decision support

## I. INTRODUCTION

Building evacuation safety is difficult to evaluate directly in operational environments because full-scale emergency drills are costly, disruptive, and limited in the range of hazardous conditions that can be reproduced safely. Designers and safety planners historically rely on simplified code-based calculations or isolated assumptions that do not adequately capture route competition, localized congestion, bottleneck formation, and sensitivity to exit availability. Simulation offers a practical alternative by enabling structured what-if analysis under repeatable conditions.

While many tools produce pedestrian trajectories, there remains a gap in reproducible, floor-plan-driven safety assessment workflows that integrate seamlessly with architectural inputs. To bridge this gap, this study introduces *PeopleFlow*, a framework emphasizing end-to-end geometry extraction and comparative evaluation over claiming absolute real-world safety guarantees.

This paper makes the following contributions:

- A reproducible floor-plan-to-simulation pipeline.
- A quantitative safety assessment framework evaluating dynamic scenarios (occupancy stress, blocked exits, routing variation).
- Interpretable safety metrics grounded in comparative evaluation across diverse real-world layouts.
- Statistical validation of evacuation performance metrics demonstrating robust capacity for smart city planning and disaster preparedness.

## II. RELATED WORK

Existing literature spans macroscopic crowd simulations, microscopic agent-based models, and building information modeling (BIM). Compared to density-aware routing and cellular automata, modern force-based methods offer high fidelity but demand significant computational overhead. Previous approaches in digital twin simulation and smart building safety typically limit testing to singular building instances, ignoring structural variability. This work acts as a decision-support framework rather than introducing a radically new pedestrian physics model, prioritizing wide scenario matrices across diverse architectures to evaluate resilience systematically.

## III. PEOPLEFLOW FRAMEWORK

The core pipeline converts floor plan images or CAD data into navigable geometry. It models exits, obstacles, and localized hazards. The framework enforces reproducibility through exact seeding and reproducible configuration artifacts. A simulation kernel handles agents utilizing varied routing preferences. Outputs are captured in real-time to report metrics dynamically.

## IV. SIMULATION METHODOLOGY

The application evaluates evacuation through physics-based pedestrian movement and parameterized routing heuristics (shortest path vs. congestion-aware).

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**Algorithm 1** PeopleFlow Simulation Loop
 

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1: Initialize agents at spawn locations
2: Initialize available exits and navigation mesh
3: while not all agents evacuated and  $t < T_{max}$  do
4:   Compute forces (social, physical, attraction)
5:   Update velocity  $\mathbf{v}$  based on forces
6:   Update position  $\mathbf{x} = \mathbf{x} + \mathbf{v}\Delta t$ 
7:   Update density map  $\rho(\mathbf{x}, t)$ 
8:   Record metrics
9: end while
  
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### A. Mathematics of Congestion

To measure congestion quantitatively, the system respects the fundamental relations of pedestrian dynamics. The flow rate  $q$  dynamically links to pedestrian density  $\rho$  and mean velocity  $v$ :

$$q = \rho \cdot v \quad (1)$$

Congestion forms when localized density peaks and restricts forward propagation velocity to near zero.

To provide statistical reliability, the framework iterates each scenario configuration  $n = 10$  times. The mean evacuation time  $\bar{T}$  and standard deviation  $\sigma$  are defined as:

$$\bar{T} = \frac{1}{n} \sum_{i=1}^n T_i, \quad \sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (T_i - \bar{T})^2} \quad (2)$$

Assumptions include a flat floor layout, primarily 2D abstraction, and simplified panic distributions, without claiming universal behavioral realism.

## V. SAFETY ASSESSMENT METHOD

Safety assessment requires multi-metric evaluation to map risk dynamically. The framework extracts total evacuation time (90% clearance target), peak crowd density, bottleneck count, and exit load balance. Safer configurations correspond directly to reductions in extreme density distributions and operational clearance timing without disproportionately loading a single exit path.

## VI. CASE STUDY DESIGN

We evaluate the framework across 9 specific floor plans, including generic corridor baseline limits, academic multi-room configurations, hospitals, hotels, and uploaded real-world mall blueprints.

For every case study, 5 standardized scenarios enforce structured variance: **Scenario 1: Normal Evacuation (Baseline)** – Typical nominal load operating under shortest-path logic.

**Scenario 2: High Occupancy** – Agents increased by 50% to simulate overcrowding limits.

**Scenario 3: Blocked Exit** – A primary exit is disabled, forcing rerouting.

**Scenario 4: Routing Policy Comparison** – Introduces congestion-aware agents avoiding high-density zones.

**Scenario 5: Panic Variability** – Introduces massive desired speed variability and erratic collision factors.

## VII. RESULTS

Table I presents the quantitative metrics resulting from 10 simulation runs for each layout-scenario configuration.

Across configurations, the introduction of a blocked exit heavily degrades baseline performance (e.g., Academic Plan clearance spikes from 182.68s down to congestion-induced slowdowns). The deployment of least-crowded routing (S4) actively drops clearance bottlenecks compared to nominal distributions across nearly all real cases.

## VIII. DISCUSSION

The extensive evaluation illustrates that the most informative safety indicator is the degradation profile across geometric designs rather than an isolated nominal estimate. The multi-layout results confirm that designs heavily reliant on few central arteries (e.g., complex multi-room academic plans) are highly brittle when placed under panic or blocked-exit stresses. Statistical variation ( $\sigma$ ) similarly rises in structurally complex plans under panic logic, heavily implicating variance in layout geometry as a primary driver of unpredictable evacuations.

### A. Limitations

Simplified human behavior limits exact behavioral mappings. The 2D operational plane ignores vertical staircase compression. Furthermore, panic behavior is modeled abstractly by increasing desired velocity variations without rich localized psychological modeling.

## IX. CONCLUSION

This paper positions evacuation simulation not as a mathematical guarantee of safety but as an actionable decision-support tool. By formalizing building safety into a reproducible computational matrix evaluating normal operation against disruption, the PeopleFlow engine provides quantitative evidence to systematically identify critical layout vulnerabilities. Future work involves extending the framework into fully calibrated 3D architectural spaces, integrating physical dynamic fire hazards, and utilizing denser localized trajectory learning sets to overcome existing deterministic assumptions.

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TABLE I  
SIMULATION RESULTS ACROSS 9 LAYOUTS AND 5 SCENARIOS (10 RUNS PER CONFIGURATION)

| Case Study       | Scenario    | Evacuation Time (s) |                  | Peak Density (agents/m <sup>2</sup> ) |                  |
|------------------|-------------|---------------------|------------------|---------------------------------------|------------------|
|                  |             | Mean ( $\bar{T}$ )  | Std ( $\sigma$ ) | Mean ( $\bar{\rho}$ )                 | Std ( $\sigma$ ) |
| Academic_Plan    | S1_Baseline | 182.68              | 16.44            | 120.49                                | 3.91             |
|                  | S2_HighOcc  | 191.92              | 9.48             | 131.74                                | 4.83             |
|                  | S3_Blocked  | 190.34              | 6.78             | 125.66                                | 3.53             |
|                  | S4_Routing  | 38.26               | 2.21             | 24.55                                 | 2.3              |
|                  | S5_Panic    | 140.14              | 32.15            | 104.46                                | 12.85            |
| Hospital_Plan    | S1_Baseline | 30.12               | 2.08             | 8.54                                  | 0.99             |
|                  | S2_HighOcc  | 32.64               | 2.45             | 12.92                                 | 1.64             |
|                  | S3_Blocked  | 46.96               | 3.49             | 18.37                                 | 2.35             |
|                  | S4_Routing  | 32.52               | 1.8              | 8.84                                  | 1.35             |
|                  | S5_Panic    | 45.5                | 4.66             | 7.98                                  | 0.82             |
| Hotel_Plan       | S1_Baseline | 138.02              | 31.96            | 100.13                                | 15.16            |
|                  | S2_HighOcc  | 121.02              | 27.47            | 80.16                                 | 15.75            |
|                  | S3_Blocked  | 153.24              | 39.78            | 110.32                                | 11.44            |
|                  | S4_Routing  | 144.64              | 28.53            | 179.94                                | 6.47             |
|                  | S5_Panic    | 131.18              | 37.19            | 57.07                                 | 2.1              |
| Library_Plan     | S1_Baseline | 52.84               | 2.97             | 21.8                                  | 2.0              |
|                  | S2_HighOcc  | 57.22               | 4.35             | 33.99                                 | 3.17             |
|                  | S3_Blocked  | 59.08               | 3.79             | 30.25                                 | 1.55             |
|                  | S4_Routing  | 58.76               | 5.23             | 22.26                                 | 1.09             |
|                  | S5_Panic    | 70.96               | 10.73            | 20.93                                 | 2.54             |
| Mall_Plan        | S1_Baseline | 24.18               | 2.43             | 8.31                                  | 1.61             |
|                  | S2_HighOcc  | 26.88               | 1.87             | 12.15                                 | 1.59             |
|                  | S3_Blocked  | 37.4                | 2.04             | 13.65                                 | 1.43             |
|                  | S4_Routing  | 35.1                | 1.67             | 9.3                                   | 1.45             |
|                  | S5_Panic    | 42.86               | 4.63             | 8.66                                  | 1.21             |
| Plan1            | S1_Baseline | 37.14               | 2.7              | 10.85                                 | 0.74             |
|                  | S2_HighOcc  | 37.58               | 0.93             | 15.98                                 | 1.77             |
|                  | S3_Blocked  | 51.26               | 4.44             | 18.18                                 | 2.15             |
|                  | S4_Routing  | 39.02               | 5.32             | 10.92                                 | 2.51             |
|                  | S5_Panic    | 46.58               | 3.88             | 9.4                                   | 1.21             |
| Plan2            | S1_Baseline | 36.62               | 3.11             | 9.3                                   | 1.85             |
|                  | S2_HighOcc  | 38.06               | 2.56             | 12.45                                 | 2.19             |
|                  | S3_Blocked  | 39.52               | 2.63             | 12.91                                 | 2.92             |
|                  | S4_Routing  | 42.26               | 3.68             | 9.7                                   | 1.93             |
|                  | S5_Panic    | 44.92               | 4.29             | 8.67                                  | 1.57             |
| Plan3            | S1_Baseline | 166.54              | 24.92            | 124.42                                | 21.07            |
|                  | S2_HighOcc  | 170.24              | 14.7             | 145.21                                | 16.0             |
|                  | S3_Blocked  | 200.0               | 0.0              | 241.19                                | 4.31             |
|                  | S4_Routing  | 86.82               | 27.08            | 187.85                                | 2.73             |
|                  | S5_Panic    | 114.26              | 11.65            | 39.08                                 | 4.76             |
| Supermarket_Plan | S1_Baseline | 25.7                | 1.15             | 11.17                                 | 1.56             |
|                  | S2_HighOcc  | 27.04               | 1.75             | 15.58                                 | 1.79             |
|                  | S3_Blocked  | 26.3                | 1.1              | 11.38                                 | 1.09             |
|                  | S4_Routing  | 29.14               | 2.39             | 9.7                                   | 1.54             |
|                  | S5_Panic    | 29.96               | 1.68             | 8.92                                  | 1.19             |

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